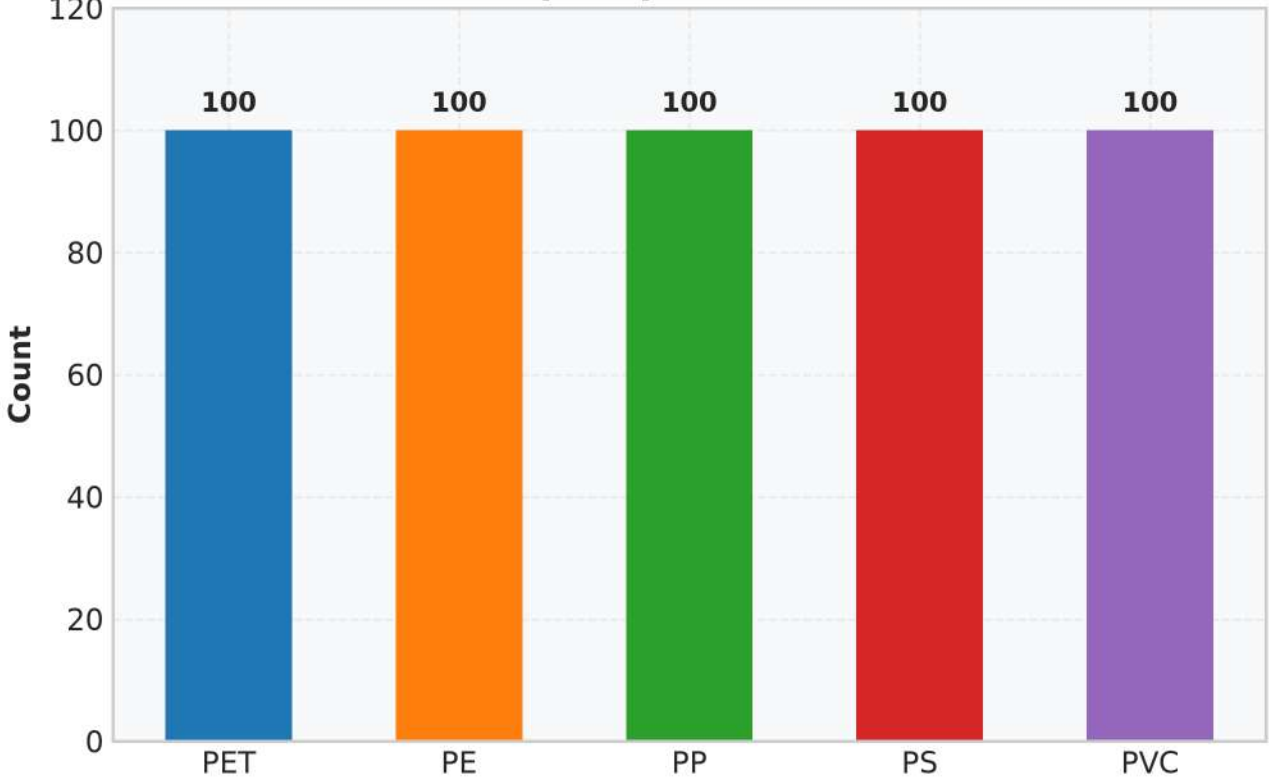
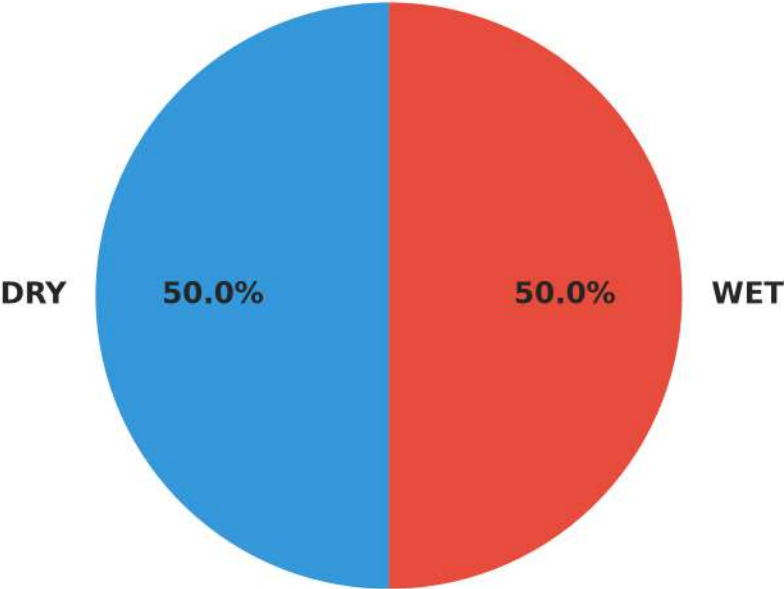


# NIR Microplastic Dataset Composition & Metadata Dashboard

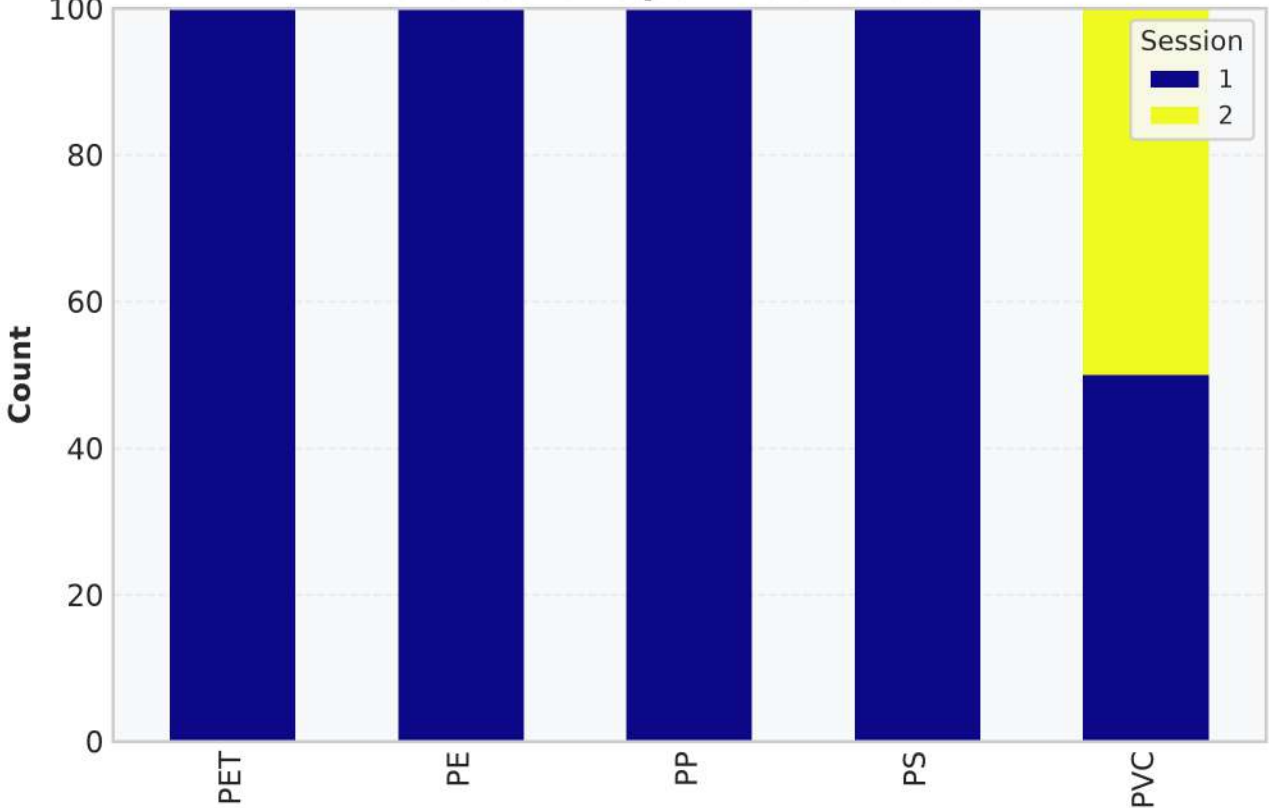
A. Samples per Plastic Class



B. DRY vs WET Condition Ratio



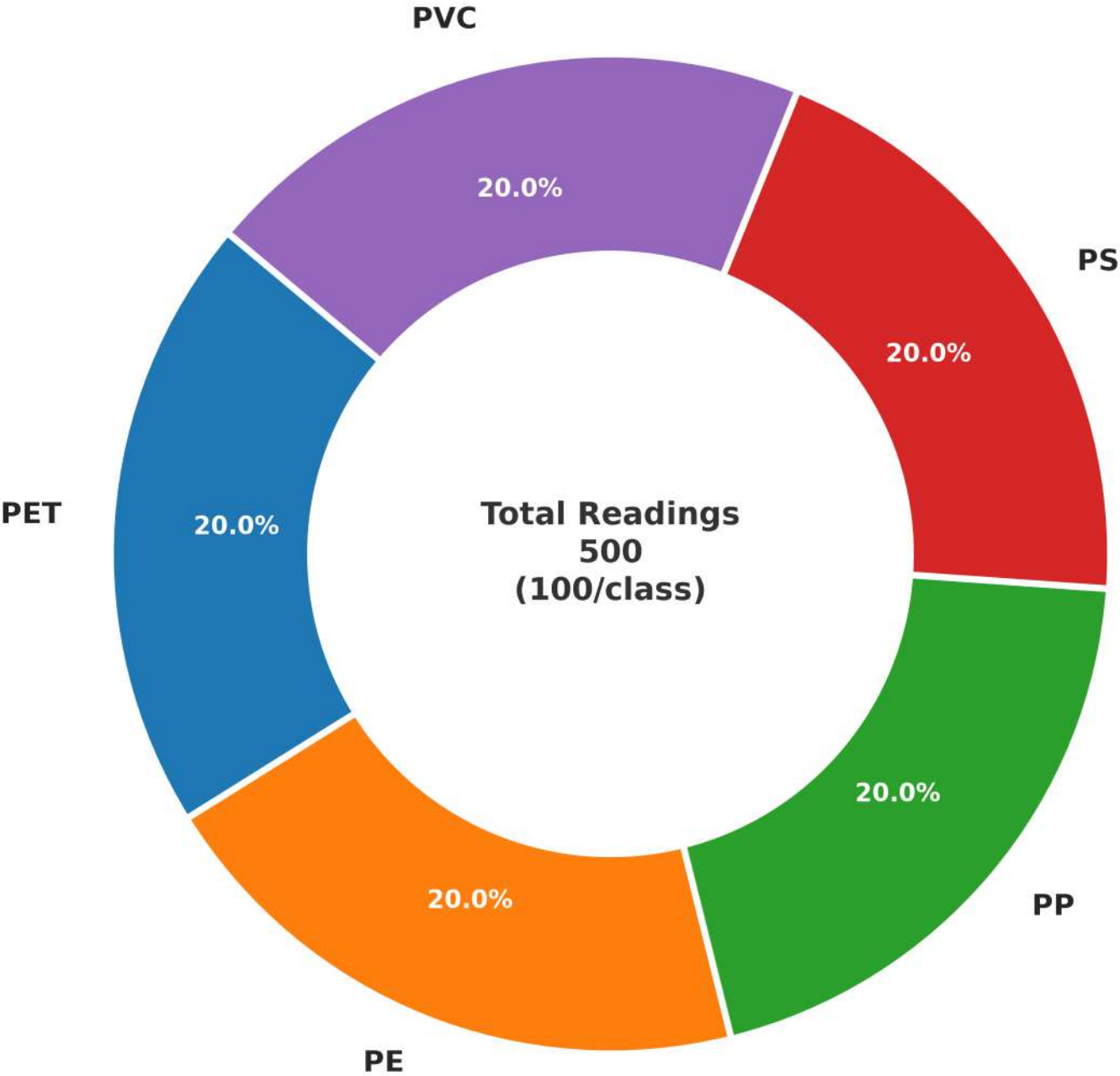
C. Sessions per Plastic Class



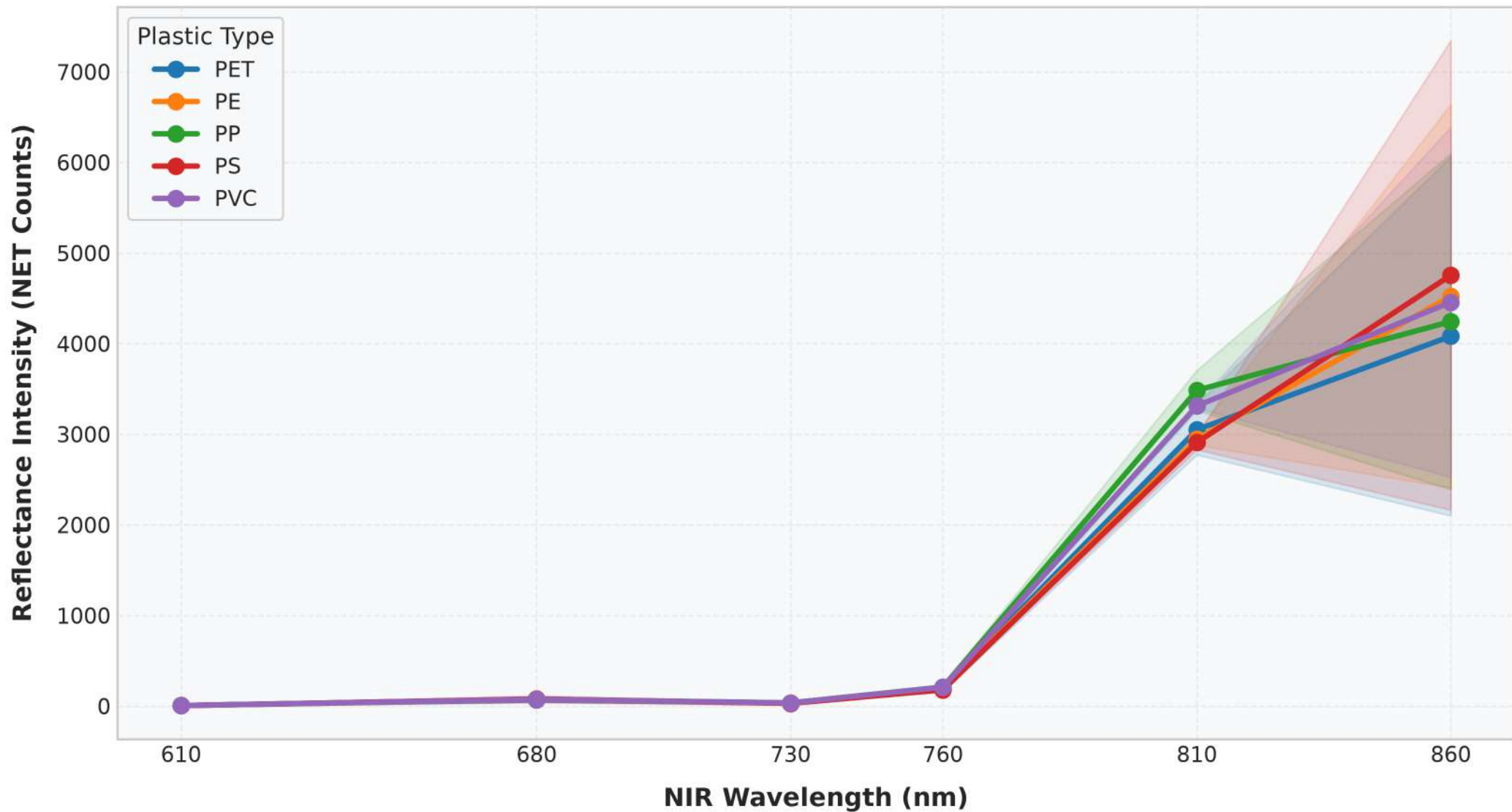
## DATASET COMPOSITION SUMMARY

- Total Calibrated Spectra: 500 rows
- Target Plastic Classes: 5 (PET, PE, PP, PS, PVC)
- Samples per Class: Exactly 100/class
- DRY Samples: 250 rows (50.0%)
- WET Samples: 250 rows (50.0%)
- Spectral Range: 6 Channels (610–860 nm)
- Raw Data Integrity: Deduplicated & Reconciled

# Dataset Balance: 5 Plastic Classes

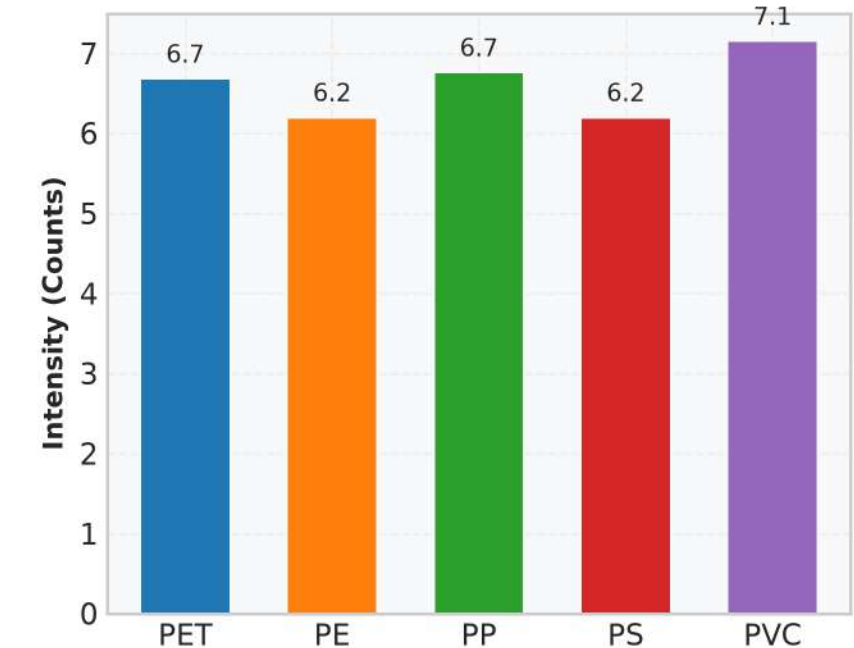


# NIR Spectral Signatures by Plastic Type (610-860 nm)

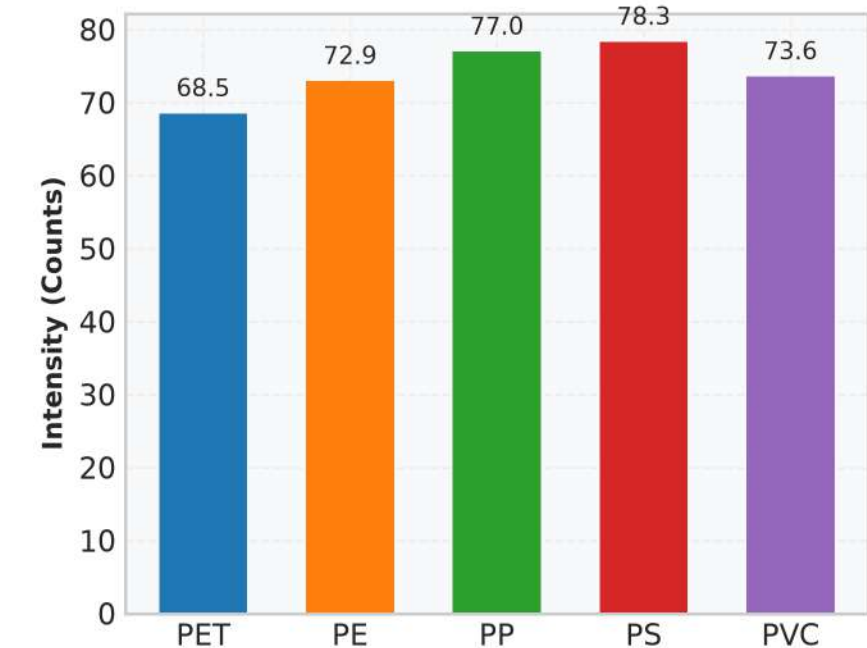


# Per-Channel Mean Intensity Comparison Across Plastics

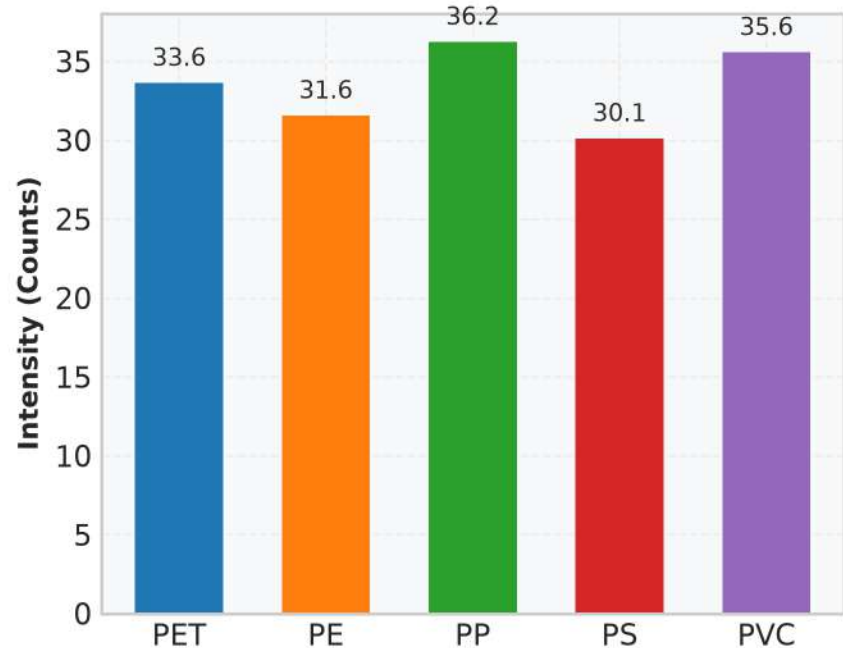
Channel 610 nm (CH610)



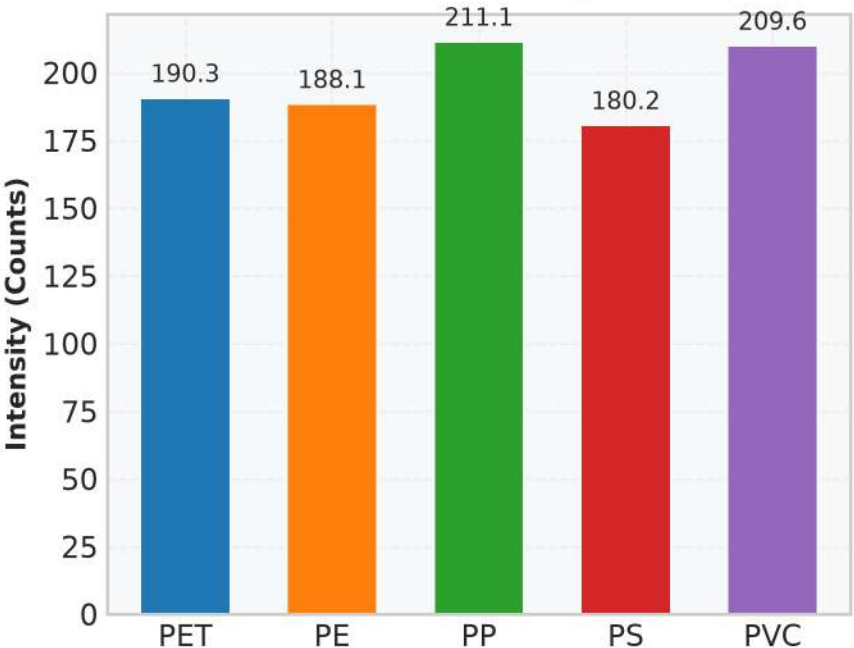
Channel 680 nm (CH680)



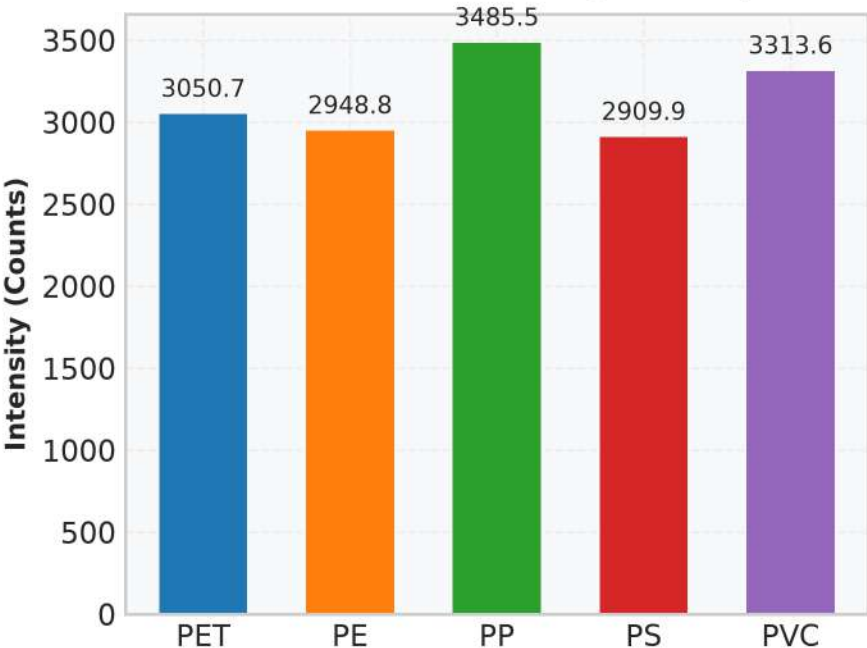
Channel 730 nm (CH730)



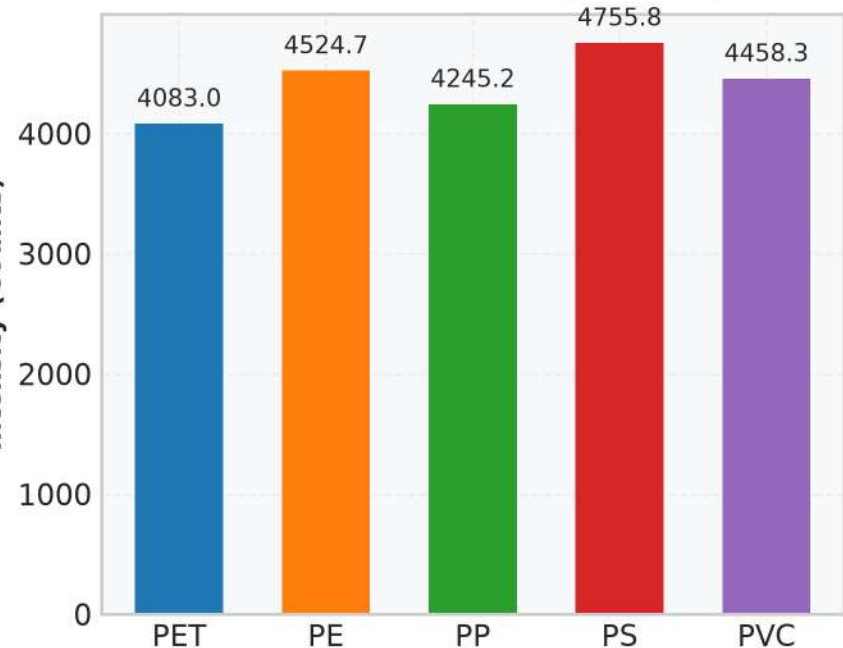
Channel 760 nm (CH760)



Channel 810 nm (CH810)

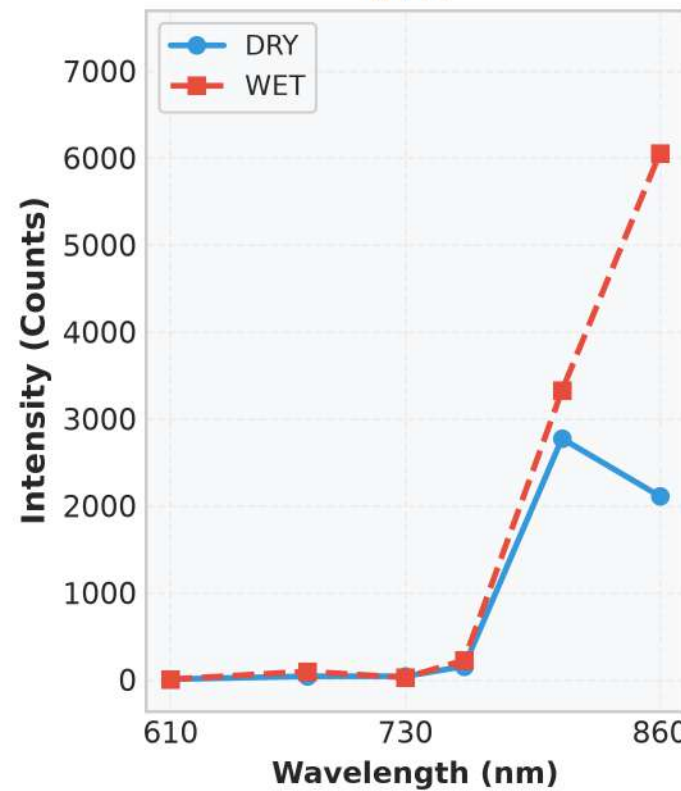


Channel 860 nm (CH860)

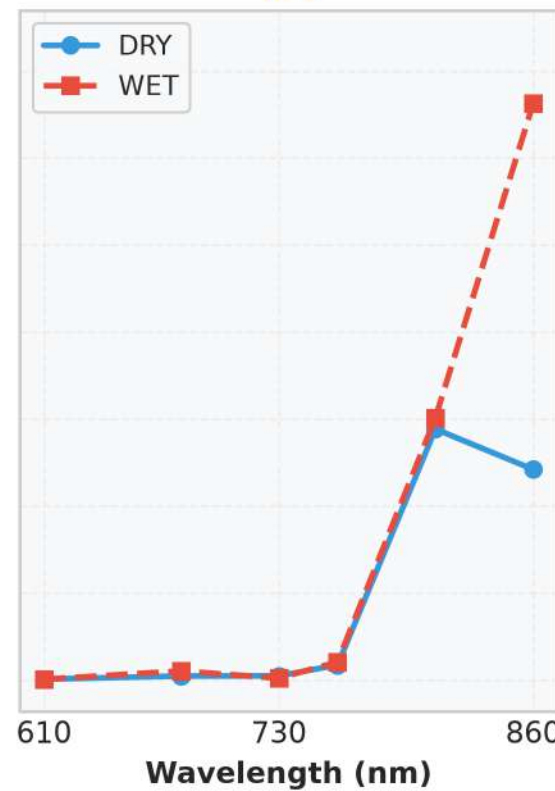


# Spectral Response Shift: DRY vs WET Sample Conditions

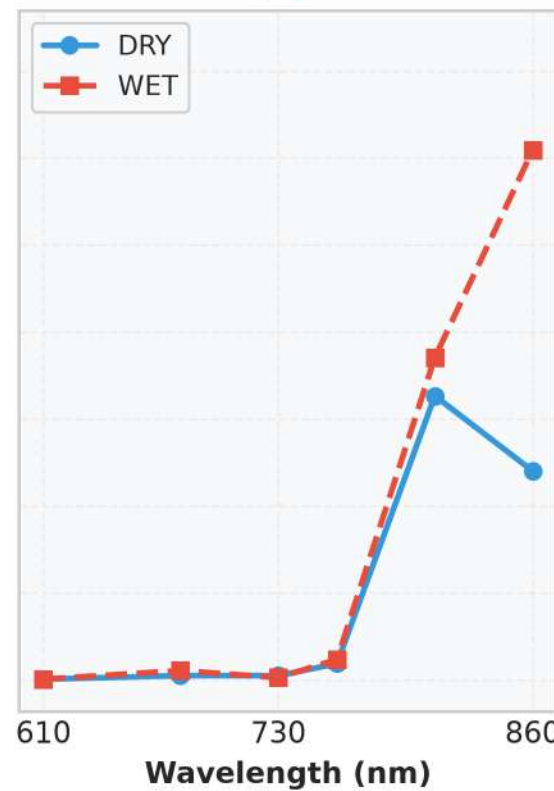
PET



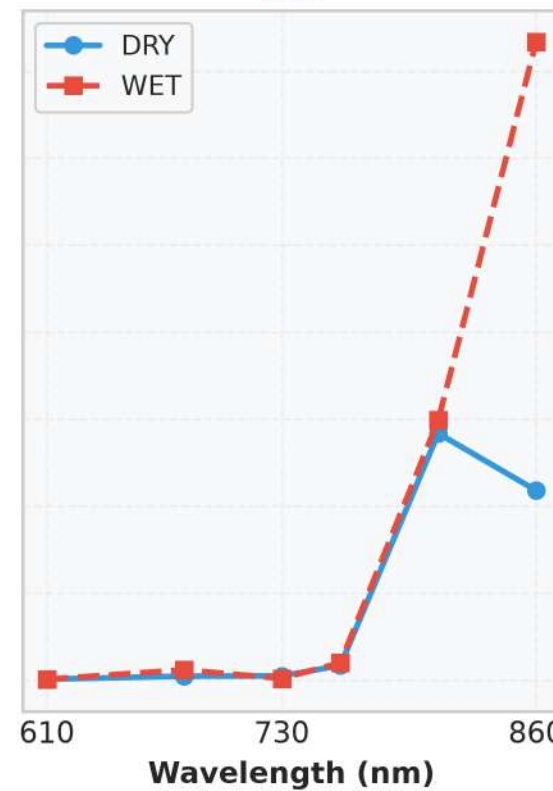
PE



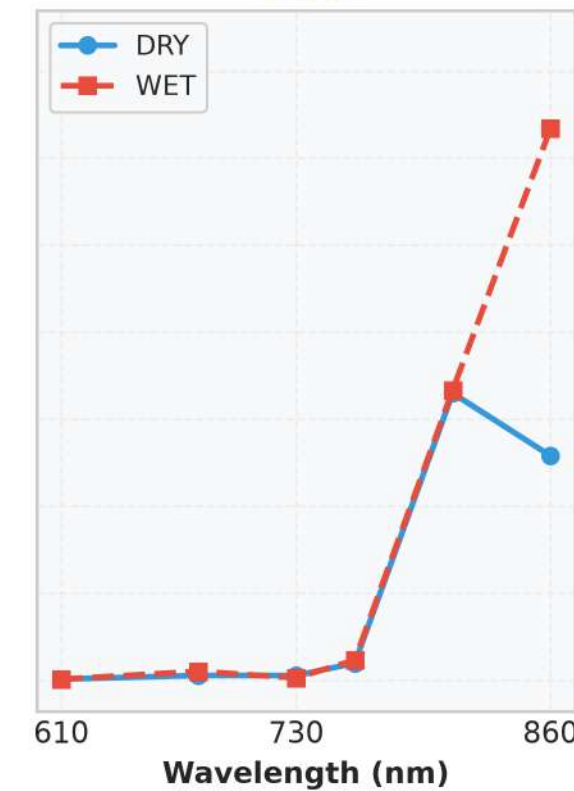
PP



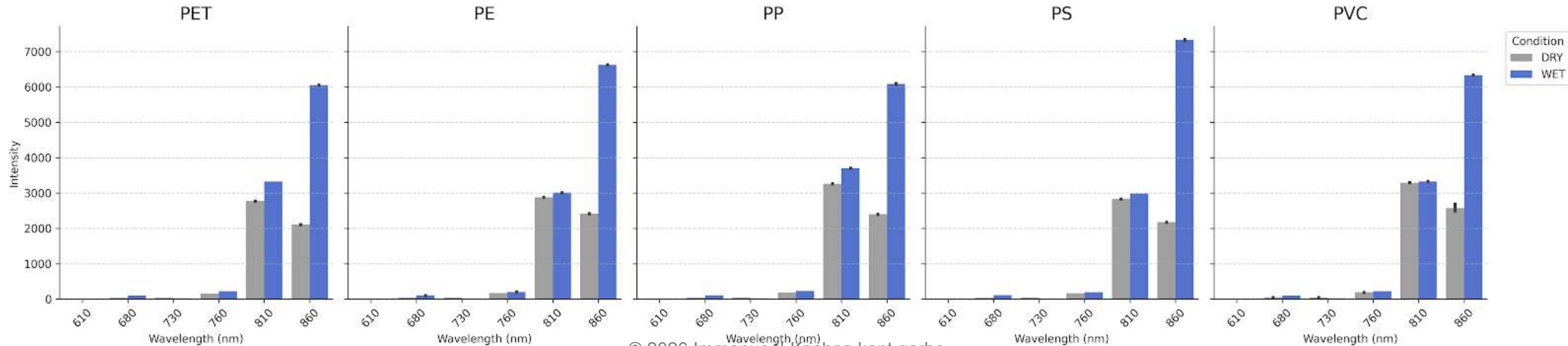
PS



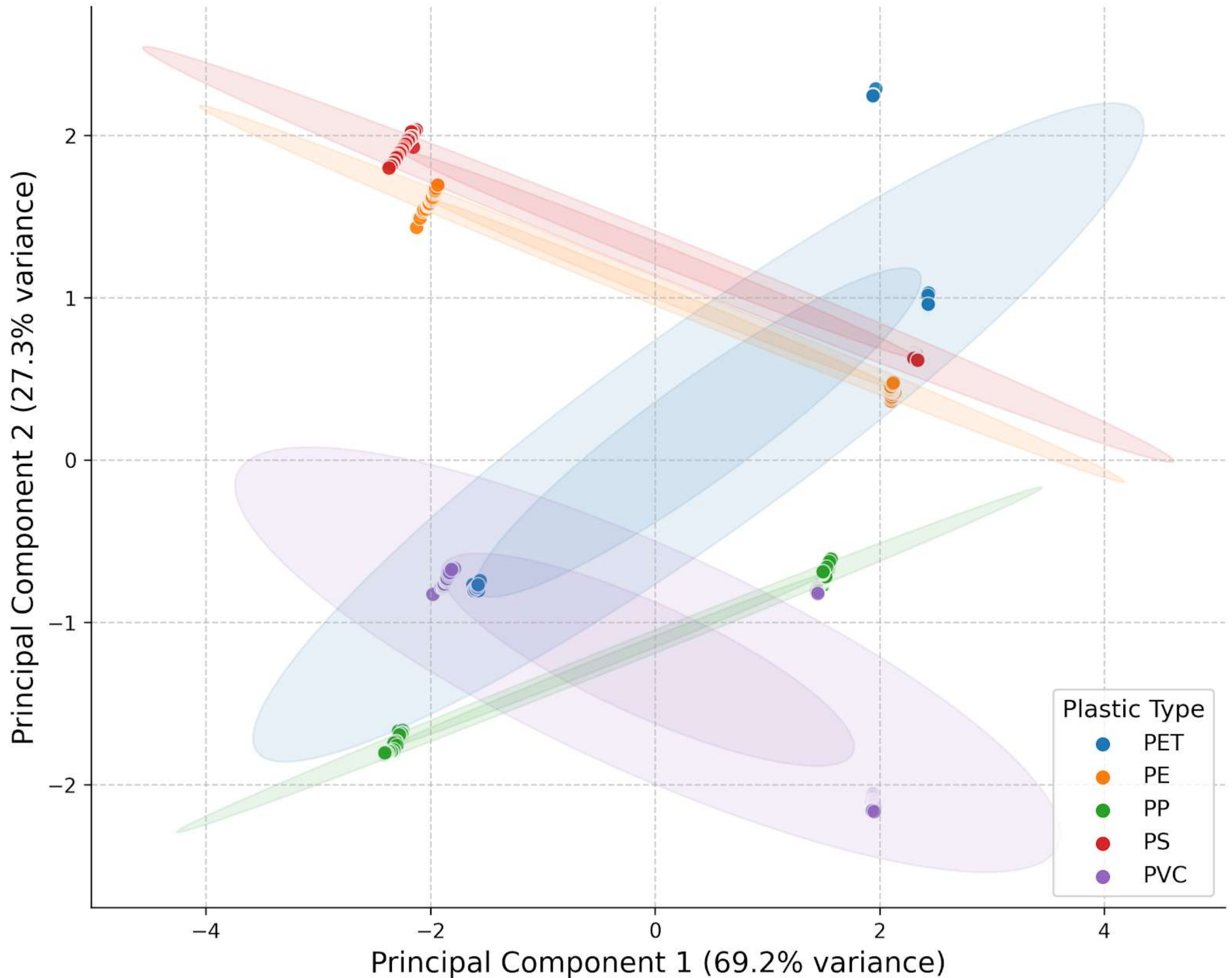
PVC



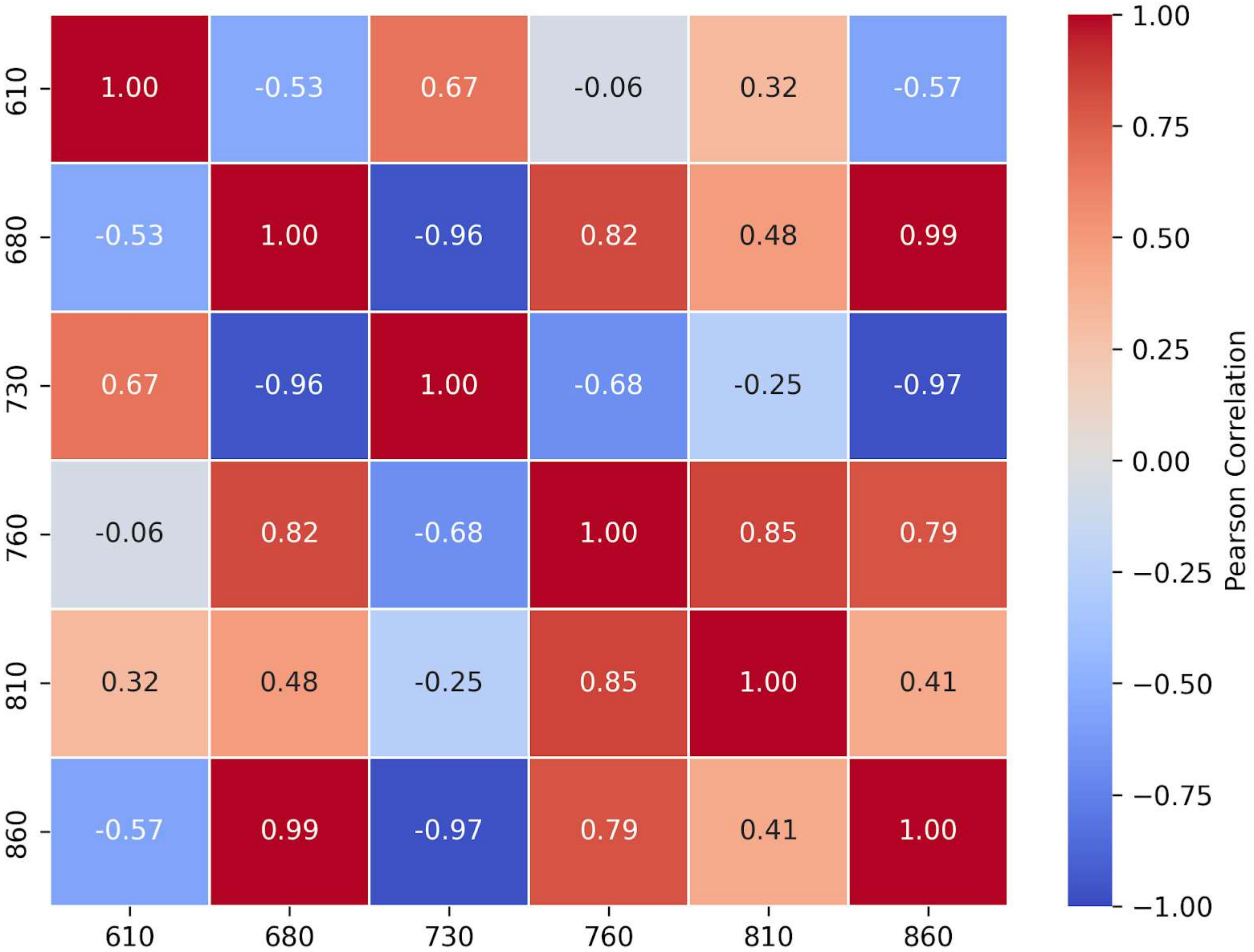
## DRY vs WET Condition Spectral Shift



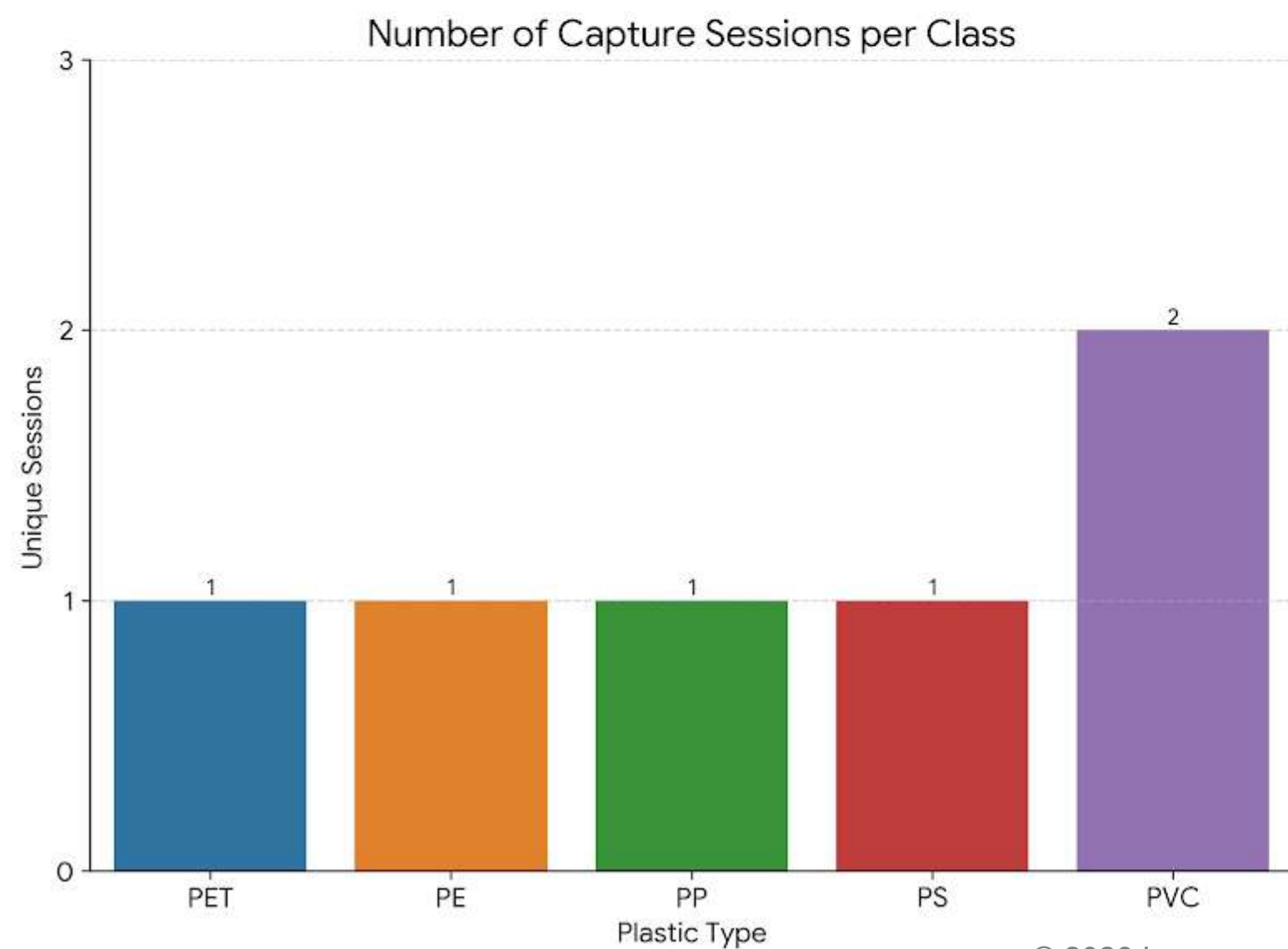
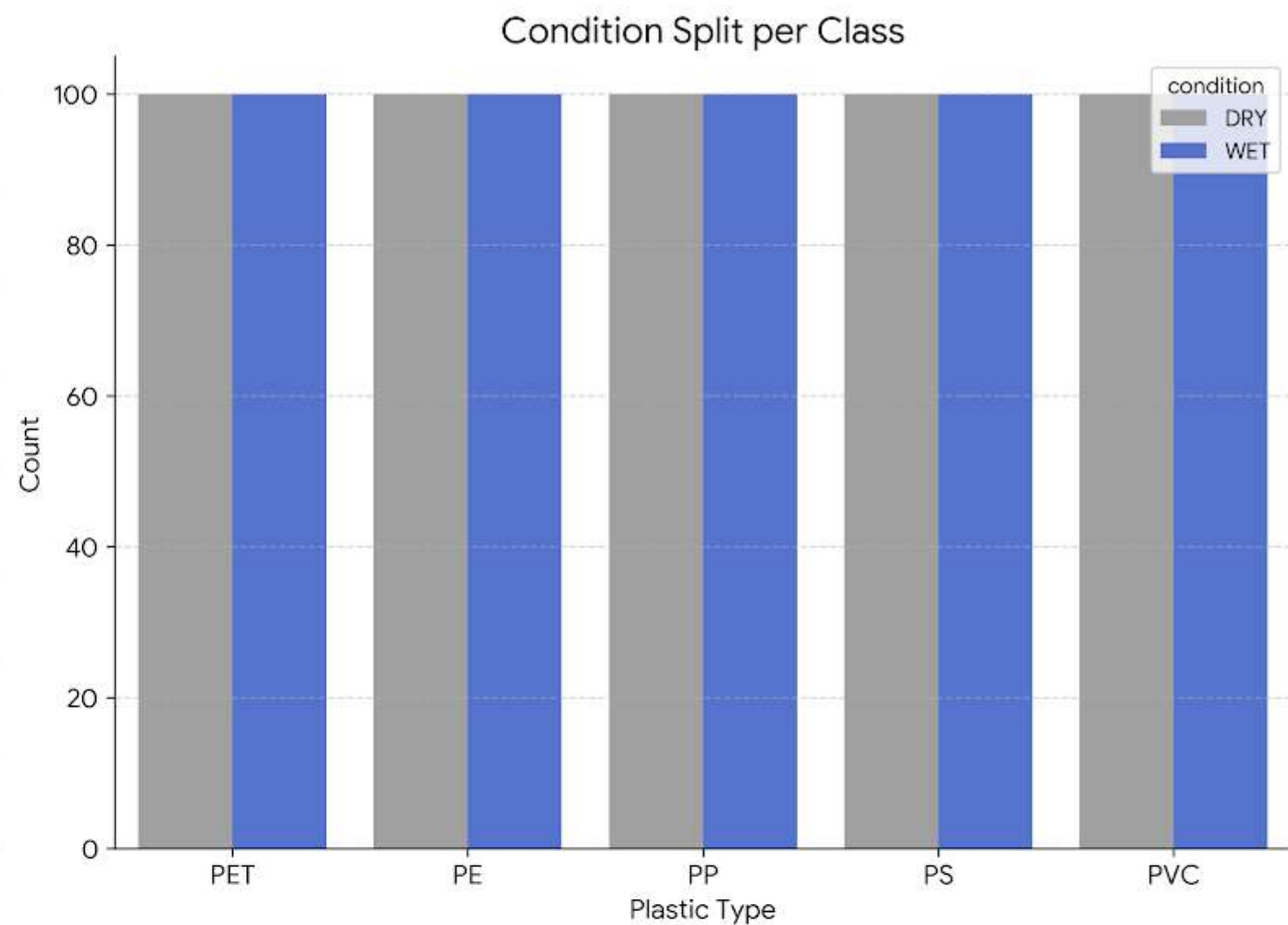
# Class Separability (PCA)



# Channel Correlation Matrix



# Dataset Composition Overview



Total Valid Samples

10000

Across 5 polymer types  
Using Phase: NIR\_ONLY

# NIR MICROPLASTIC CLASSIFIER

Machine Learning Performance Metrics & Deployment Status

**97.0%**

**RBF-SVM Classification Accuracy  
(5 Target Plastic Classes)**

**5-Fold CV**

**Cross-Validation Evaluation  
(Standard Normal Variate SNV)**

## **VALIDATION CAVEAT & DEPLOYMENT NOTICE**

*Accuracy metrics were evaluated on single-session laboratory dataset captures.  
Multi-session validation across environmental samples, varied geometries, and field conditions is required.*